

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Conditions for life on Earth - Set 1

Mark scheme

AQA spec 3.1.1; The living environment; textbook pp. 5-9

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark each 1 mark for each of the 3 points below. 1 Mars has low mass so weak gravity and could not retain a dense atmosphere 1 Very low atmospheric pressure means water would boil / evaporate away even at low temperatures 1 Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • insolation → incoming solar radiation / energy arriving from the Sun	3

Question	Marking guidance	Marks
01.2	1 mark each 1 mark for each of the 3 points below. 1 Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO₂, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil 1 Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured 1 Too close (Venus) - too hot – too far (Mars) - too cold Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. insolation → incoming solar radiation / energy arriving from the Sun	3
01.3	1 mark each 1 mark for each of the 2 points below. 1 The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface 1 The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions) Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	1 mark each 1 mark for each of the 2 points below. 1 The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun 1 Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted – the Moon's gravity keeps the tilt stable and creates tides Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. insolation → incoming solar radiation / energy arriving from the Sun	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Question	Marking guidance	Marks
02.1	1 mark each 1 mark for each of the 4 points below. 1 General physiological solvent: most biological reactions take place between substances dissolved in water 1 Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes 1 High specific heat capacity moderates temperature changes – evaporation absorbs heat (cooling) 1 Anomalous expansion: ice floats so water bodies do not freeze solid – absorbs UV protecting early marine life – provides aquatic habitats Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Anomalous → an outlier / a result that does not fit the pattern Do not credit - Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. - discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	4
02.2	1 mark each 1 mark for each of the 4 points below. 1 From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen – at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) 1 Oxygen in the stratosphere absorbed UV, forming ozone (O₃) in a dynamic equilibrium of O, O₂ and O₃ 1 The ozone layer reduced UV reaching the surface so organisms could survive out of the water 1 Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved – many anaerobic archaea died out Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. ozone layer → the ozone in the stratosphere that absorbs UV Do not credit - answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4
02.3	1 mark each 1 mark for each of the 2 points below. 1 Photosynthesis removed CO₂ (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels 1 This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. infrared → heat radiation / long-wave radiation re-emitted by the Earth	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
03.1	1 mark each 1 mark for each of the 3 points below. 1 Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion – waste products of one process are raw materials of another so wastes do not build up 1 Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere 1 Allowing more plant growth, further stabilising climate and soils Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. transpiration → water lost as vapour from the leaves	3
03.2	Any 3 of 4 1 mark each for any 3 of the 4 points below, in any order, maximum 3. 1 Lack of ancient historical / instrumental data 1 Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions 1 Lack of sophisticated equipment / inability to measure many factors 1 Limited coordination between researchers – lack of data collection in many areas (oceans, poles) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. reliability → repeatable / gives consistent results when repeated Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
03.3	1 mark each 1 mark for each of the 4 points below. 1 The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell 1 Air bubbles trapped in ice give the composition of the past atmosphere (CO₂, methane) 1 Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) 1 More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	
04.1	Correct answer only 1 mark for the correct option. No mark if more than one box is ticked. 1 A: Its mass	1

Question	Marking guidance	Marks
04.2	1 mark each 1 mark for each of the 2 points below. 1 The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced 1 The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. insolation → incoming solar radiation / energy arriving from the Sun	2
04.3	1 mark each 1 mark for each of the 2 points below. 1 The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze 1 Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away – at much lower pressure there would be no liquid water Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
05.1	Calculation 1 mark for the correct answer; working does not have to be shown. 1 $2.7 - 2.45 = 0.25$ billion = 250 million years Also accept - The correct answer however it is set out, including a different but valid method. - An answer carried forward from an earlier incorrect step (error carried forward). - An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. - The value alone - the unit (million years) is already printed on the answer line. Do not credit - A bare wrong answer with no working.	1
05.2	1 mark each 1 mark for each of the 2 points below. 1 For millions of years all the oxygen produced reacted with (dissolved) iron in the oceans 1 Only when all the iron had reacted did surplus dissolved oxygen build up in the oceans and then escape into the atmosphere Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
05.3	Any 3 of 4 1 mark each for any 3 of the 4 points below, in any order, maximum 3. 1 Oxygen in the atmosphere absorbed ultraviolet light 1 A dynamic equilibrium of reactions between O, O₂ and O₃ (the ozone layer) developed 1 Less UV reached the surface so organisms were no longer restricted to water (which had absorbed the UV) 1 Aerobic respiration became possible so more complex / larger organisms (plants and animals) evolved – many anaerobic Archaea and bacteria died out Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. ozone layer → the ozone in the stratosphere that absorbs UV Do not credit - answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
05.4	1 mark each 1 mark for each of the 2 points below. 1 Photoautotrophs absorbed carbon dioxide, a greenhouse gas, and some of the carbon was stored in geological sediments such as carbonate rocks and fossil fuels (carbon sequestration) 1 The lower CO₂ concentration reduced the retention of heat, offsetting the increase in the brightness of the Sun (about 10% per billion years) Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.5	Any 2 of 3 1 mark each for any 2 of the 3 points below, in any order, maximum 2. 1 Biogeochemical cycles developed so small amounts of nutrient elements could support life for long periods without depletion 1 Transpiration by land plants returned water vapour to the atmosphere and increased rainfall in other areas 1 Regulation of atmospheric composition (eg oxygen / CO₂ by photosynthesis and respiration) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Transpiration → water lost as vapour from the leaves Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-8; AQA spec 3.1.1.2.	
06.1	1 mark for the point below. 1 Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Question	Marking guidance	Marks
06.2	1 mark each 1 mark for each of the 4 points below. 1 Age / date of the sample (radioisotope dating) 1 Past temperatures 1 Composition of the past atmosphere (eg CO₂ / methane concentrations) 1 Temperature: more magnesium is incorporated into calcite at higher temperatures Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	4
06.3	1 mark each 1 mark for each of the 2 points below. 1 Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present 1 Ocean currents redistribute heat and nutrients – changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	Any 3 of 4 1 mark each for any 3 of the 4 points below, in any order, maximum 3. 1 There was no oxygen / ozone layer so high levels of UV radiation reached the land surface – water absorbed the UV and protected organisms in the oceans 1 Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely 1 The atmosphere contained toxic gases such as ammonia – the early Archaea were anaerobic so did not need oxygen 1 Water acted as the solvent for reactions / transport Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. ozone layer → the ozone in the stratosphere that absorbs UV Do not credit - Points after the first 3 - the question asks for 3, so a list of extras cannot pick up the marks. - answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3.	